Nikola Izzan Ar Rasheed

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SUMMARY

Computer Engineering graduate (GPA 3.72/4.00) with experience in data analysis, machine learning, and NLP through an internship at the Financial Services Authority (OJK), Bank Negara Indonesia (BNI), and various AI projects. Built an end-to-end NLP pipeline (BERTopic, SVM, Random Forest, XGBoost; >80% accuracy) on 6,000+ financial reports at OJK, deployed Qwen2.5-VL on NVIDIA Jetson Orin Nano for a thesis project, and built a transformer-based semantic recommendation system (MiniLM, IndoBERT) with TensorFlow.

EDUCATION

University of Brawijaya | Malang, East Java, Indonesia — August 2022 - July 2026

- Computer Engineering | GPA: 3.72 / 4.00

WORK EXPERIENCES

Data Management and Statistics Department — Financial Services Authority (OJK) — June 2025 - July 2025

Data Analyst Intern — (2 Months)

- Built an end-to-end NLP pipeline (preprocessing, BERTopic clustering, topic labeling, evaluation) to classify 6,000+ financial institution reports.
- Trained and compared SVM, Random Forest, and XGBoost classifiers, achieving >80% accuracy.
- Analyzed financial reports using Python and Excel and developed automated data processing and reporting workflows using Python and Excel VBA.

IT Strategy & Architecture Division — Bank Negara Indonesia (BNI) — September 2026 - present

IT & Digital Operations Intern

- Participated in business–IT alignment meetings involving BNI business users, technology vendors, and IT stakeholders, gaining exposure to IT project planning and requirements discussions. Documented SDLC processes across BNI's IT divisions as process diagrams.

PROJECT EXPERIENCE

Smart Cane with Audio-Visual Description System for the Visually Impaired — August 2025 – December 2025

(Undergraduate Thesis)

- Built an on-device assistive system on NVIDIA Jetson Orin Nano 8GB that turns camera images into Indonesian speech: Qwen2.5-VL 3B (via Ollama)→offline EN→ID translation (ArgosTranslate) → Piper-TTS.
- Evaluated on 50 real images, each with 5 reference captions (2 human, 3 LLM-generated): average METEOR 0.45, end-to-end latency 24.1 s, peak power 25.5 W (~1.5 h estimated runtime on a battery pack).

TravelMate.AI — AI Travel Chatbot — May 2025 – July 2025

- Built a semantic recommendation system for East Java destinations, hotels, and restaurants using MiniLM and IndoBERT embeddings with cosine-similarity ranking and location filtering; fine-tuned MiniLM and tuned IndoBERT-based models with Keras Tuner in TensorFlow/Keras.
- Scraped and cleaned datasets (90 hotels, 67 restaurants) and collaborated in a team of 6 to integrate the model with backend and frontend.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Machine Learning / AI: VLM, OpenCV, scikit-learn, TensorFlow, Keras, HuggingFace

Data & Visualization: Excel, PostgreSQL, Pandas, Seaborn, NumPy, Matplotlib, Streamlit

Tools: Git, GitHub, Linux, Ollama, Figma

CERTIFICATIONS, TRAINING, AND ACHIEVEMENTS

- **English Proficiency (2026):** BEPT — Score: 553/677
- **Training (2025):** Coding Camp by DBS Foundation — Machine Learning Engineer
- **Achievement (2024):** 2nd Place Winner, Business Plan Competition — Technoday UNNES 2024
- **Course (2025):** Data Analysis with Python, Data Processing Fundamental, Machine Learning Development, Applied Machine Learning — Dicoding Indonesia